Daryl Jay S. Alidro **Midterm Activity #3 – Basic ATM System**

BSIT 2-3 **April 17, 2026**

Program:

```java
package mypackage;
import java.util.Scanner;

public class BasicATMSystemAlidro {
    static Scanner input = new Scanner(System.in);

    public static void displayMenu() {
        System.out.println("===== ATM MENU =====\r\n"
                + "1. Check Balance\r\n"
                + "2. Deposit\r\n"
                + "3. Withdraw\r\n"
                + "4. Exit\r\n");
    }

    public static void checkBalance(float balance) {
        System.out.println("Balance: " + balance + "\n");
    }

    public static float deposit(float balance) {
        float deposit = 0;

        do {
            System.out.println("Input deposit amount: ");
            deposit = input.nextFloat();
            if (deposit < 0) {
                System.out.println("Invalid Amount. Try Again");
            } else {
                System.out.println("Processing...\r\n"
                        + "Done!\n");
                break;
            }
        } while (deposit < 0);

        return balance += deposit;
    }

    public static float withdraw(float balance) {
        float withdraw = 0;
```

```java
    public static float withdraw(float balance) {
        float withdraw = 0;

        do {
            System.out.println("Input withdraw amount: ");
            withdraw = input.nextFloat();
            if (withdraw < 0) {
                System.out.println("Invalid Amount. Try Again");
            } else if (withdraw > balance) {
                System.out.println("Insufficient Balance. Try Again");
            } else if (withdraw > 2000) {
                System.out.println("Maximum withdrawal is ₱2000 only.\r\n"
                        + "Try Again.");
            } else {
                System.out.println("Processing...\r\n"
                        + "Done!\n");
                break;
            }
        } while (withdraw < 0 || withdraw > balance || withdraw > 2000);

        return balance -= withdraw;
    }

    public static void main(String[] args) {
        int choice;
        float balance = 5000;

        do {
            displayMenu();
            System.out.println("Choice: ");
            choice = input.nextInt();

            switch (choice) {
            case 1:
                checkBalance(balance);
                break;
            case 2:
                balance = deposit(balance);
```

```java
            } else {
                System.out.println("Processing...\r\n"
                        + "Done!\n");
                break;
            }
        } while (withdraw < 0 || withdraw > balance || withdraw > 2000);

        return balance -= withdraw;
    }

    public static void main(String[] args) {
        int choice;
        float balance = 5000;

        do {
            displayMenu();
            System.out.println("Choice: ");
            choice = input.nextInt();

            switch (choice) {
            case 1:
                checkBalance(balance);
                break;
            case 2:
                balance = deposit(balance);
                break;
            case 3:
                balance = withdraw(balance);
                break;
            case 4:
                System.out.println("Exiting....\r\n"
                        + "Thank you!");
                break;
            default:
                System.out.println("Invalid Input. Try Again.\n");
            }
        } while (choice != 4);
    }
}
```

User Input & Output:

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Choice:
2
Input deposit amount:
5000
Processing...
Done!

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Choice:
1
Balance: 10000.0

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Choice:
3
Input withdraw amount:
5000
Maximum withdrawal is ₱2000 only.
Try Again.
Input withdraw amount:
2000
```

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Choice:
1
Balance: 8000.0

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
```

```
Choice:
2
Input deposit amount:
-5000
Invalid Amount. Try Again
Input deposit amount:
2000
Processing...
Done!

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Exit

Choice:
4
Exiting....
Thank you!
```